LPMF — Presentation Script

Load Profile Management and Forecasting

Slide-by-slide speaker notes in Hinglish, with formulas, worked examples and viva Q&A

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Poori kahani 1 minute mein

Agar examiner kahe 'ek minute mein batao project kya hai', to yeh bolo (~60 second):

Delhi ki bijli ki demand har saal garam, peaky aur mehngi hoti ja rahi hai. 29 June 2026 ko Delhi ka peak 8,748 MW tak gaya. DISCOM ko har din agle din ki demand 96 blocks mein, yaani har 15 minute ke liye, batani padti hai, aur galti hone par CERC ke DSM rules ke under paisa lagta hai. Humne ek dashboard banaya jo sirf forecast nahi dikhata, balki uska nateeja bhi dikhata hai. Real Delhi SLDC load data (April 2023 se January 2026) aur Open-Meteo weather ko saaf kiya, 15-minute blocks banaye, aur ek LightGBM model train kiya jo agle din ke 96 blocks predict karta hai. Test period par iski error (MAPE) 3.93% hai, jabki simple seasonal-naive baseline ki 5.16%. Har forecast ke saath ek P10–P90 band bhi hai, jisme 84.4% actual values aayi, target 80% tha. Dashboard ke saare 12 pages live hain: chhe core forecast-pipeline par (Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator, Transformer Health), aur chhe additional engines jo apne-apne real ya foreign benchmark se check hue hain (IEX Price, ToD Bill Simulator, Solar & Net Load, Losses & PF, Theft & Segments, Data & Model Card). Har number par label hai ki woh REAL hai, SYNTHETIC hai, MODELLED hai, FOREIGN hai ya ILLUSTRATIVE. Aur hum apni limitations bhi khul ke batate hain: Diwali jaise din par error zyada hai, test set mein garmi ka peak season nahi hai, aur Solar Plant 2 ka fit genuinely weak hai, jiska root cause investigate kiya gaya.

Timing table

Slide	Title	Time (sec)	Clock (end)
1	Title	35	0:35
2	Problem Statement	75	1:50
3	Gap and USP	70	3:00
4	Dashboard: 12 of 12 Live	55	3:55
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6	Model Accuracy & Fit Check	75	6:05
7	Overview & Load Analytics	100	7:45
8	Weather Impact & Forecast	105	9:30
9	DSM Calculator & Transformer Health	95	11:05
10	IEX Price & ToD Bill Simulator	95	12:40
11	Solar & Net Load, and Losses & PF	100	14:20
12	Theft & Segments, and Data & Model Card	90	15:50
13	Benchmarks	50	16:40
14	References	25	17:05
15	Thank You	25	17:30
	Total	1050	17:30 (≈ 17.5 min)

Speed ka andaaza: ~140 shabd per minute. Viva ke liye 3–5 minute alag rakho. Agar time kam ho, slide 5 (Dataset), slide 6 (Accuracy) aur slide 8 (Weather Impact & Forecast) poori bolo; baaki ki script chhoti kar sakte ho.

Slide 1 — Title

Slide par kya hai

Project ka naam (Load Profile Management & Forecasting), Faculty of Technology, University of Delhi, team ke chaar naam aur dono guides: Dr. Ujjal Sur (Supervisor) aur Dr. Arjun Tyagi (Co-Supervisor).

Bolne wali script (~35 sec)

Good morning sir/ma'am. Hum aaj apna project present kar rahe hain: Load Profile Management and Forecasting, yaani Delhi grid ke liye ek AI-integrated load intelligence dashboard. Main hoon Vansh

Sachan, aur mere saath hain Animish Gupta, Ayush Rathore aur Lavish Rawat. Yeh project humne Dr. Ujjal Sur aur Dr. Arjun Tyagi ki guidance mein kiya hai. Agle kuch minute mein hum batayenge ki problem kya hai, humne kya banaya, aur har number kahan se aaya.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Project aur team ka introduction.

Kyu kiya: Audience ko pehle hi pata chal jaye ki topic power system aur machine learning dono ka hai.

Kaise kiya: Ek line ka subtitle: 'AI-integrated load intelligence dashboard for the Delhi grid'.

Viva mein pooche ja sakte hain

Q: Project ka one-line summary kya hai?

A: Real Delhi data se agle din ki 96-block load forecast, aur us forecast ko peak alert, DSM ₹ aur transformer ageing jaise engineering nateejon tak le jaana.

Q: Isme AI kahan hai?

A: Forecast LightGBM model se aata hai, jo ek machine learning model hai. Baaki pages engineering formulas hain, except Theft & Segments jahan dusra ML model (Isolation Forest, K-means, logistic regression -> LightGBM) chalta hai.

Agli slide par: *To chaliye, pehle dekhte hain ki Delhi ki demand ki problem itni serious kyun hai.*

Slide 2 — Problem Statement

Slide par kya hai

Chaar bade numbers: India ka all-time peak 256.1 GW, Delhi ka 8,748 MW, heat index se 67% daily-peak variation (CSE 2025), aur IEX price ₹1.11 se ₹9.71 per unit. Neeche problem aur chaar asar: penalty, transformer heating, steep evening, losses.

Bolne wali script (~75 sec)

Delhi ki demand garam, peaky aur galat hone par mehngi hai. 25 April 2026 ko India ka peak 256.1 GW tha, aur 29 June 2026 ko Delhi ka record 8,748 MW. CSE ki 2025 study kehti hai ki Delhi ke daily peak ke variation ka lagbhag 67% heat index se samjha ja sakta hai. Heat index matlab temperature aur humidity ko milake 'kitni garmi mehsoos hoti hai'. DISCOMs ko agle din ki demand 96 blocks of 15 minutes mein batani padti hai. Agar forecast galat ho, to CERC DSM Regulations 2024 ke under har MW deviation ka charge lagta hai. Saath hi, 40 degree ke din full load par transformer ki insulation lagbhag 10 guna tez umar kha jaati hai, solar ki wajah se shaam ki demand tezi se chadhti hai, aur AT&C losses abhi bhi 15.04% hain. Aaj ke monitoring tools sirf bata dete hain ki kya ho chuka; yeh nahi batate ki kal ki galti ka kya nateeja hoga.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Problem ko numbers ke saath define kiya.

Kyu kiya: Sirf 'forecast karna hai' kehne se importance samajh nahi aati; paisa, equipment aur grid par asar dikhana zaroori tha.

Kaise kiya: Har number ke saath official source likha: PIB, Delhi SLDC, CSE, EAC-PM, IEC 60076-7, 14th Integrated Rating.

Formula / number ka matlab

96 blocks ka matlab

1 din = 24 ghante × 4 blocks per ghanta = 96 blocks, har block 15 minute ka. DSM settlement isi block level par hota hai, isliye forecast bhi 96 blocks ka chahiye.

Viva mein pooche ja sakte hain

Q: 67% wala number aapka hai?

A: Nahi, yeh CSE 2025 ka hai aur daily peak ke liye hai. Hamara apna fit har 15-min block par $r^2 = 0.61$ deta hai. Dono alag metric hain, isliye directly compare nahi karte.

Q: 8,748 MW aapke data mein hai?

A: Nahi. Hamara data 12 January 2026 par khatam hota hai, aur yeh peak 29 June 2026 ka hai. Data ka maximum 8,581.6 MW (19 June 2024) hai. Isliye 8,748 MW sirf alert limit ki tarah use kiya.

Agli slide par: *Ab sawal yeh hai ki baaki projects yeh problem kyun solve nahi karte; yeh agli slide mein.*

Slide 3 — Gap and USP

Slide par kya hai

Do column: typical load-forecasting projects kya karte hain, aur hamara system kya alag karta hai. Neeche USP box.

Bolne wali script (~70 sec)

Zyadatar forecasting projects ek graph dikha ke ruk jaate hain, jaise 'kal 8,000 MW'. Ek model, ek accuracy number, aur aksar foreign ya toy dataset. Hum forecast ko uske nateeje tak le jaate hain. Pehla, model ladder: har complex model ko apne se simple model ko beat karna padta hai, warna hum use nahi rakhte. Aur hum imaandari se batate hain ki h = 1–2, yaani 15–30 minute aage, simple persistence abhi bhi jeet jaata hai. Doosra, data real hai: Delhi SLDC ka 5-minute data aur Open-Meteo weather. Teesra, har forecast ke saath P10–P90 band hai, aur selected din par SHAP explanation. Chautha, DSM ₹ aur transformer life ke what-if calculators. Hamara USP: design ke hisaab se ek Delhi forecast chhe engines chalata hai, aur ab ToD bill, solar ramp, losses aur theft bhi live hain – sirf forecast nahi, har engine ship ho chuka hai. Alag-alag tools GitHub aur Kaggle par milte hain, lekin unhe Indian regulation par ek saath jodna naya hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Existing kaam aur hamare kaam ka farak saaf kiya.

Kyu kiya: Examiner ka pehla sawal hota hai 'isme naya kya hai?'.

Kaise kiya: Paanch point side-by-side compare kiye; sabhi 12 engines ab live hain, koi future scope baaki nahi.

Viva mein pooche ja sakte hain

Q: Persistence kya hai?

A: Sabse simple forecast: agla block = abhi wala block. 15 minute mein load bahut kam badalta hai, isliye bahut chhote horizon par yeh bahut accurate hota hai.

Q: SHAP kya hai?

A: Ek method jo batata hai ki kis feature ne forecast ko kitna upar ya neeche dhakela. Jaise 'kal ka load' sabse bada driver nikla.

Agli slide par: *Ab dekhte hain ki yeh sab dashboard par kaise dikhta hai.*

Slide 4 — Dashboard: 12 of 12 Live

Slide par kya hai

12 cards, sab rangeen aur 'LIVE'. Pehle chhe (Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator, Transformer Health) core forecast-pipeline par; baaki chhe (IEX Price, ToD Bill Simulator, Solar & Net Load, Losses & PF, Theft & Segments, Data & Model Card) apne-apne benchmark ke against validate hue hain.

Bolne wali script (~55 sec)

Dashboard mein 12 pages hain, aur har page ek operational sawal ka jawab deta hai. Saare baarah live hain aur real API se chalte hain. Overview batata hai kal ka peak kitna aur kab hoga. Load Analytics batata hai demand kitni peaky hai. Weather Impact batata hai garmi se kitna load badhta hai. Forecast mein 96 blocks, uncertainty band aur model comparison hai. DSM ₹ Calculator aur Transformer Health what-if calculators hain. Baaki chhe pages ab bhi live hain: IEX Price real historical exchange price use karta hai, ToD Bill Simulator real DERC rates, Solar & Net Load real plant data se validate hua, Losses & PF real PFC AT&C figures ke saath, Theft & Segments real SGCC benchmark par, aur Data & Model Card har page ka summary deta hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Poore system ka map ek slide par.

Kyu kiya: Sabhi engines ko live dikhana imaandaari hai, lekin har engine ka apna benchmark bhi batana zaroori hai.

Kaise kiya: Next.js frontend, FastAPI backend; har card provenance badge (REAL, SYNTHETIC, ILLUSTRATIVE, MODELLED, FOREIGN) ke saath.

Viva mein pooche ja sakte hain

Q: Data & Model Card ka data kahan se aata hai?

A: Yeh ek static summary page hai, docs/MODEL_RESULTS.md aur API responses se banaya gaya hai -- koi naya data nahi, sirf har page ka source aur result ek jagah.

Q: Live pages ka data kahan se aata hai?

A: FastAPI se. Har response mein provenance (REAL / ILLUSTRATIVE / MODELLED / FOREIGN / SYNTHETIC) aur source string hoti hai.

Agli slide par: *Har page ka number kahan se aata hai, yeh agli slide batati hai: dataset.*

Slide 5 — Dataset & Variables

Slide par kya hai

Table: dashboard ke original chhe features, har ek ka source, key variables (exact field names), REAL ya ILLUSTRATIVE tag, aur cleaning ka kaaran. Neeche synthetic data ka note, ab Theft & Segments ke demo fleet tak simit.

Bolne wali script (~55 sec)

Yeh slide batati hai ki har number kahan se aata hai. Pehle chaar features, yaani Overview, Load Analytics, Weather Impact aur Forecast, poori tarah REAL data par chalte hain: Delhi SLDC ka load, April 2023 se January 2026, aur Open-Meteo weather Delhi ke coordinates par. Variables wahi hain jo code mein hain, jaise load_mw, heat_index_c, temp_c aur rh_pct. Cleaning causal hai, yaani har decision sirf past data dekh kar hota hai: Hampel outlier filter, gap filling, aur train, validation aur test ka chronological split. DSM ₹ Calculator aur Transformer Health data-driven nahi hain; user value deta hai aur formula chalta hai, isliye unpar hamesha ILLUSTRATIVE likha hai. Synthetic data, jaise residential ya EV load profiles, ab sirf Theft & Segments ke demo fleet (page 11a) ke liye hai; baaki sab pages real ya clearly-labelled illustrative data par chalte hain.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Har feature ka source, variables aur tag ek table mein.

Kyu kiya: Engineer ko pata hona chahiye ki kis number par kitna bharosa karein; real aur illustrative ko mix karna khatarnaak hai.

Kaise kiya: Field names seedhe app/schemas.py aur engine functions se liye; har chart par provenance badge.

Formula / number ka matlab

Row counts (DATA_QUALITY.md, data files)

5-minute grid par 293,184 slots chahiye the, 285,252 mile (2.705% missing). 15-minute par 97,728 blocks, jinme se 1,330 (1.361%) khaali chhode gaye, kyunki 3 ghante se lambe gap fill nahi karte. Usable blocks = 97,728 − 1,330 = 96,398.

Viva mein pooche ja sakte hain

Q: Causal cleaning kyun zaroori hai?

A: Wahi load series features aur targets dono banati hai. Agar cleaning future dekhe, to future ki jaankari features mein leak ho jaati hai aur score jhootha achha aata hai.

Q: Gap filling kaise kiya?

A: Gap ki lambai ke hisaab se: chhote gap forward-fill (894 blocks), medium gap pichhle 5 din ke same time ke profile se (1,153 blocks), aur 3 ghante se lambe gap khaali chhode.

Agli slide par: *Model kitna accurate hai, aur kya woh overfit ya underfit hai? Yeh agli slide batati hai.*

Slide 6 — Model Accuracy & Fit Check

Slide par kya hai

Left: train / validation / test MAPE ka chart (model vs seasonal-naive) aur teen accuracy tiles. Right: MAPE, accuracy, naive MAPE aur gain ki table; horizon-wise MAPE; overfitting / underfitting ka verdict; aur do honest caveats. Tag: REAL.

Bolne wali script (~75 sec)

Ab dekhte hain ki model kitna accurate hai, aur kya woh overfit ya underfit hai. Accuracy hum 100 minus MAPE se define karte hain. Test par MAPE 3.93% hai, yaani accuracy lagbhag 96%. Train par MAPE 2.29%, accuracy 97.71%; validation par 4.29%, accuracy 95.71%. Seasonal-naive, yaani kal isi block ka load, train par 5.44%, validation par 5.97% aur test par 5.16% deta hai, isliye model test par 24% kam galti karta hai. Overfitting check: train aur test ke beech gap hai, lekin chhota hai. Early stopping 3,000 mein se sirf 436 rounds par ruki, aur validation aur test error flat rahi; sirf train error giri. Isliye overfitting nuksaan wali nahi hai. Underfitting bhi nahi hai, kyunki model har horizon par naive se behtar hai: h = 1–4 par 3.21%, h = 5–24 par 3.48%, h = 25–48 par 3.84% aur h = 49–96 par 4.22%, jabki naive lagbhag 5.16%. Do honest caveats: pehla, test set poori tarah untouched nahi tha, kyunki target-aligned lags pehle model ki test errors dekh kar design hue, isliye 3.93% thoda optimistic ho sakta hai. Doosra, sabse chhote horizon par persistence model se behtar hai: h = 1 par 1.44% vs 3.18%.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Saved model ka train, validation aur test par accuracy aur overfitting / underfitting check.

Kyu kiya: Sirf ek accuracy number kaafi nahi; examiner poochhega ki model ratta to nahi maar gaya (overfit) ya bahut simple to nahi (underfit).

Kaise kiya: Saved model ko bina retrain kiye har split ke saare origin × horizon rows par score kiya; seasonal-naive se compare kiya; learning curve aur early-stopping rounds dekhe.

Formula / number ka matlab

Accuracy = 100 − MAPE; Gain = 1 − MAPE(model) / MAPE(naive)

Test: 100 − 3.93 = 96.07% ≈ 96%. Train 97.71%, validation 95.71%.

Gain: 1 − 3.93 / 5.16 = 24% (validation 28%, train 58%).

Early stopping: 436 rounds, limit 3,000 (patience 50).

Viva mein pooche ja sakte hain

Q: Overfitting hai ya nahi?

A: Train–test gap chhota hai (2.29% vs 3.93%), early stopping 436/3000 par ruki, aur validation aur test error flat hain, sirf train girti hai. Isliye overfitting nuksaan wali nahi hai.

Q: 96% accuracy kaise nikali?

A: Accuracy = 100 − MAPE. Test MAPE 3.93% hai, to 100 − 3.93 = 96.07% ≈ 96%. Yeh classification accuracy nahi hai; har block ki galti ko actual load se divide karke average li gayi.

Q: Validation test se zyada kyun?

A: Validation ka do-tihai hissa March–June ki garmi hai, aur naive bhi wahan zyada galti karta hai (5.97% vs 5.16%); isliye woh window mushkil hai.

Agli slide par: *Ab ek-ek karke feature-pair slides dekhte hain, shuru karte hain Overview aur Load Analytics se.*

Slide 7 — Overview & Load Analytics

Slide par kya hai

Do column: left Overview (kal ka peak, time, alert) aur formula; right Load Analytics (load factor, PAR, duration curve) aur formula. Dono REAL.

Bolne wali script (~100 sec)

Is slide par do pages hain. Pehla, Overview: yeh batata hai kal ka peak kitna hoga, kab hoga, aur kya woh limit cross karega. Formula simple hai: 96 forecast blocks mein sabse bada value peak hai. Hamare test ke aakhri din, 12 January 2026 ka example: peak block 42 mein, time 10:15 IST, peak 5,670.83 MW, jo Delhi ke record 8,748 MW se kam hai, isliye alert Clear hai, 3,077 MW headroom. Doosra, Load Analytics: load factor, yaani average ko peak se divide karna, aur PAR iska ulta. Poore period mein, April 2023 se January 2026, average 4,316 MW aur peak 8,582 MW, to LF = 0.503 aur PAR = 1.99. Load-duration curve saare blocks ko bade se chhote load tak sort karta hai; left mein tez spike dikhti hai, matlab sabse zyada load bahut kam time ke liye aata hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Agle din ka peak + alert, aur poore period ki peakiness ek saath.

Kyu kiya: Operator ko limit cross hone se pehle warning chahiye, aur PAR capacity planning ke liye zaroori hai.

Kaise kiya: Dono LightGBM ke forecast/measured load par chalne wale simple engines hain (ml/src/lpmf/engines/overview.py, load_analytics.py).

Formula / number ka matlab

Overview: $\hat{P}peak = \max_t \hat{P}(t)$; alert agar $\hat{P}peak > Plimit$

Load Analytics: $LF = Pavg / Pmax$; $PAR = 1 / LF$

Viva mein pooche ja sakte hain

Q: Limit 8,748 MW hi kyun?

A: Yeh Delhi ka all-time record hai, isliye ek natural upper limit hai. Code mein limit ek parameter hai, DISCOM apni limit daal sakta hai.

Q: 0.646 aur 0.503 mein kaunsa sahi hai?

A: Dono sahi hain, lekin alag cheez hain. 0.646 ek illustrative din ka example hai (530/820); 0.503 poore data ka measured value hai (96,398 blocks). Dashboard 0.503 dikhata hai.

Agli slide par: *Peak aur peakiness dekh li; ab garmi ka asar aur agle din ka pura forecast dekhte hain.*

Slide 8 — Weather Impact & Forecast

Slide par kya hai

Left: Weather Impact (scatter + OLS line, β = 112.0 MW/°C, r^2 = 0.61). Right: Forecast (96-block, P10–P90 band, MAPE 3.93%, SHAP). Dono REAL.

Bolne wali script (~105 sec)

Left: Weather Impact. Har ek degree heat index badhne par load kitna badhta hai, yeh OLS se nikala: β = 112.0 MW per degree Celsius, r^2 = 0.61, matlab 61% variation heat index se explain hota hai — yeh CSE ke 67% (daily peak ka) se alag metric hai. Right: Forecast, project ka dil. LightGBM ek hi model se agle din ke saare 96 blocks predict karta hai. Poore test set, 13 July 2025 se 12 January 2026, par MAPE 3.93% hai, seasonal-naive ka 5.16%, yaani 24% kam galti. Har forecast ke saath P10–P90 band hai, ek 80% range; test par 84.4% actuals ismein aaye. SHAP batata hai ki kal ka load hi sabse bada driver hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Heat sensitivity aur poora 96-block forecast ek saath.

Kyu kiya: Heat sabse bada driver hai, aur DSM charge forecast error par lagta hai — dono accurate aur imaandaar uncertainty chahiye.

Kaise kiya: heat_sensitivity engine har baar data se β dobara nikalta hai; forecast multi-horizon LightGBM (37 features, Optuna-tuned) + MAPIE split-conformal band + TreeSHAP.

Formula / number ka matlab

Weather: $P = \alpha + \beta \cdot HI$; $\Delta P \approx \beta \cdot \Delta HI$

Forecast: $MAPE = (100/n)\ \Sigma\ |y - \hat{y}|\ /\ y$; $Coverage = N(P10 \le y \le P90)\ /\ n$

Viva mein pooche ja sakte hain

Q: Graph mein curve U-shape jaisi kyun hai, line seedhi kyun?

A: Thandi mein heating ki wajah se load thoda badhta hai, isliye neeche U jaisa dikhta hai. Humne ek simple linear fit rakha; isliye r^2 0.61 hai, 1 nahi.

Q: Sabse important forecast feature kaunsa nikla?

A: tlag_1day, yaani kal isi block ka load: 66.5% gain importance. Ise shuffle karne par MAPE 3.93% se 23.4% ho jaata hai, matlab yeh genuine signal hai, leak nahi.

Agli slide par: *Forecast mein galti hogi hi; ab uski keemat (DSM) aur garmi mein equipment ki umar dekhte hain.*

Slide 9 — DSM ₹ Calculator & Transformer Health

Slide par kya hai

Left: DSM ₹ Calculator (sliders + formula + ₹3.475 lakh/hr example). Right: Transformer Health (hot-spot slider + ageing curve, 10.08×). Dono ILLUSTRATIVE.

Bolne wali script (~95 sec)

DSM ka matlab Deviation Settlement Mechanism: agar DISCOM schedule se zyada ya kam bijli le, to charge lagta hai. Formula: volume limit tak ka deviation normal rate NR par, limit se upar wala hissa k guna rate par. Example: deviation 60 MW, limit 41 MW, NR ₹5,000 per MWh, k = 1.5. Total ₹86,875 per block, yaani ₹3.475 lakh per hour. Transformer Health: IEC 60076-7 ke hisaab se 98 degree hot-spot par ageing rate 1 hai; har 6 degree badhne par yeh double ho jaata hai. Example: 40 degree ambient, full load par hot-spot = 40 + 55 + 23 = 118 degree, to V = 2^((118−98)/6) = 10.08×. Matlab 1 ghanta 118 degree par chalane se 10.08 ghante ki insulation life khatam hoti hai. Dono what-if calculators hain, kisi real meter ya transformer ka measured data nahi.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Forecast deviation ko rupees mein, aur hot-spot temperature ko insulation ageing mein badalna.

Kyu kiya: Kuch MW ki galti ya kuch degree ki garmi abstract lagti hai; rupees aur ageing multiplier mein dikhane se keemat samajh aati hai.

Kaise kiya: dsm_penalty aur transformer_ageing engines, dono user ke what-if input par chalte hain, output hamesha is_illustrative = True.

Formula / number ka matlab

DSM: $C_b = \min(|D|, VL) \cdot NR/4 + \max(|D| - VL, 0) \cdot k \cdot NR/4$

Transformer: $V = 2^{((\theta h - 98)/6)}$; $L = \Sigma\, Vn \cdot \Delta t$

Viva mein pooche ja sakte hain

Q: DSM ka number real penalty kyun nahi?

A: CERC 2024 mein normal rate grid frequency se judi hoti hai; woh tables encode nahi kiye. Humne ek flat rate aur ek multiplier maana hai, isliye sirf charge ki shape dikhti hai.

Q: 55 aur 23 kahan se aaye?

A: IEC 60076-7 ke distribution transformer (ONAN) defaults: top-oil rise 55 K aur hot-spot gradient 23 K. Rated load par yeh ambient se 78 K upar le jaate hain.

Agli slide par: *Ab dekhte hain do naye engines jo isi forecast par chalte hain: real exchange price aur real tariff.*

Slide 10 — IEX Price & ToD Bill Simulator

Slide par kya hai

Left: IEX Price (real historical block-of-day price × day-ahead load). Right: ToD Bill Simulator (flat vs ToD bill, real DERC rates). Dono REAL.

Bolne wali script (~95 sec)

IEX Price: humne Kaggle se real Indian Energy Exchange ka day-ahead market data liya, 2018 se 2024 tak, 233,760 rows. Har block-of-day (din ke 96 slots) ka average price nikala aur usse day-ahead load ke saath multiply karke purchase cost banaya. Yeh real historical price range ₹3.05 se ₹6.13 per unit deta hai, jo exchange ki apni published range se match karta hai. ToD Bill Simulator: DERC ka Tariff Schedule FY2020-21, clause 3(d), seedha PDF se padha — peak hours (14:00–17:00 aur 22:00–01:00) mein 20% surcharge, off-peak (04:00–10:00) mein 20% rebate, sirf May se September. Flat bill vs ToD bill vs load shift karne ke baad ka bill, teeno dikhaye.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Forecast ko real market price aur real regulatory tariff se jodna.

Kyu kiya: Sirf load forecast kaafi nahi; DISCOM ko yeh bhi pata hona chahiye ki woh load kitne mein kharidega, aur consumer ka bill kitna shift ho sakta hai.

Kaise kiya: IEX ke liye real historical average (ek specific din ka live price nahi); ToD ke liye DERC ka primary PDF directly fetch karke parse kiya, pehle ek unverified figure correct karte hue.

Formula / number ka matlab

IEX: cost = Σ P_load(t) · 0.25h · price(t)

ToD: bill = Σ(h=0 to 23) kWh(h) · rate · m(h), jahan m(h) peak/off-peak multiplier hai

Viva mein pooche ja sakte hain

Q: IEX price kaise verify kiya?

A: Humara number IEX ki apni published range ke saath compare kiya — match karta hai, lekin yeh ek 6.7-saal ka typical-day average hai, kisi specific din ka live price nahi.

Q: ToD rates DERC se seedhe aaye?

A: Haan. Pehle ek unverified 30%/25% figure use kiya tha; fir directly DERC ke FY2020-21 tariff schedule PDF se clause 3(d) padh kar 20%/20% correct kiya.

Agli slide par: *Ab solar aur losses dekhte hain — dono mein real plant aur real regulator data use hua hai.*

Slide 11 — Solar & Net Load, and Losses & PF

Slide par kya hai

Left: Solar & Net Load (real Open-Meteo GHI, 2 real plants se validate). Right: Losses & PF (PF/I^2R what-if + real Delhi-DISCOM AT&C). Solar MODELLED, Losses ILLUSTRATIVE+REAL benchmark.

Bolne wali script (~100 sec)

Solar & Net Load: formula simple hai, GHI (irradiance) ko capacity aur performance ratio se multiply karo. Isse do real Indian solar plants ke generation data (Kaggle) se bhi validate kiya. Plant 1 par formula bahut achha fit hua, daytime $r^2 = 0.97$. Plant 2 par daytime r^2 sirf 0.21 nikla — humne chaar cheezein check ki: timestamp shift, inverter clipping, sensor placement aur din-wise pattern. Koi fixable bug nahi mila; asli wajah hai ek hi weather sensor jo 22 alag-alag inverters ke liye use hota hai, aur Nasik ke monsoon onset ke time cloud variability badh jaati hai — yeh real-world data ki seema hai, formula ki galti nahi. Losses & PF: PF-correction kVAR aur I^2R loss illustrative what-if hain, lekin saath mein real Delhi-DISCOM AT&C loss figures hain: BRPL 6.58%, BYPL 7.84%, TPDDL 5.91%, FY2023-24, seedhe PFC ke primary PDF se, Annexure 1.8, page 76.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Solar output ko real plant data se validate karna, aur illustrative losses calculator ke saath real AT&C benchmark dikhana.

Kyu kiya: Formula ko blindly trust nahi karte — real data par test karte hain, aur weak fit ko chhupane ke bajaye root-cause investigate karte hain.

Kaise kiya: Solar formula validate_against real plant CSVs (Kaggle); Losses ke AT&C numbers PFC ke primary PDF se seedhe nikale, secondary source se re-confirm kiye.

Formula / number ka matlab

Solar: $P_{solar}(t) = (GHI(t)/1000) \cdot C_{kWp} \cdot PR$; $P_{net} = P_{load} - P_{solar}$

Losses: $Qc = P(\tan\phi 1 - \tan\phi 2)$; $Ploss = 3I^2R$

Viva mein pooche ja sakte hain

Q: Solar Plant 2 ka r^2 kam kyun hai?

A: Chaar hypotheses test ki: timestamp misalignment (shift=0 hi sabse achha fit deta hai), inverter clipping (AC/DC ratio sabhi 22 inverters mein flat ~0.975, koi clipping signature nahi), sensor placement (ek hi weather sensor 22 physically-spread inverters ko cover karta hai — sabse badi wajah), aur din-wise pattern (worst din Nasik ke monsoon onset ke aas-paas cluster karte hain, patchy cloud variability ka sign). Koi fixable bug nahi mila, isliye $r^2 = 0.208$ hi final rakha.

Q: BYPL ka AT&C loss kitna hai?

A: 7.84%, FY2023-24, PFC ke primary PDF 'Report on Performance of Power Utilities 2023-24' (Annexure 1.8, page 76) se seedha. Isi table se BRPL 6.58% aur TPDDL 5.91% bhi re-confirm hue.

Agli slide par: *Ab aakhri do pages: theft detection aur ek summary data/model card.*

Slide 12 — Theft & Segments, and the Data & Model Card

Slide par kya hai

Left: Theft & Segments (logistic regression → LightGBM, real SGCC benchmark). Right: Data & Model Card (static summary of all 12 pages). FOREIGN aur REAL.

Bolne wali script (~90 sec)

Theft & Segments: humein Delhi ya kisi Indian DISCOM ka public consumer-level smart-meter dataset with theft labels nahi mila, isliye humne real SGCC dataset use kiya — China State Grid ka public benchmark, Zheng et al. 2018 paper, 42,372 consumers, 8.53% theft rate. Logistic regression baseline se LightGBM tak gaye, test AUC-ROC 0.7585 mila, paper ke ~0.78 se thoda kam — koi leakage red flag nahi, kyunki hamara number paper se neeche hai, upar nahi. Ek demo fleet bhi hai jo synthetic consumer profiles par anomaly detection dikhata hai, sirf illustration ke liye. Data & Model Card: yeh ek static summary page hai — har ek baarah pages ka data source, provenance aur result ek table mein, taaki koi bhi number ka trace mil sake.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Theft detection ek real foreign benchmark par, aur sab kuch ek summary page mein.

Kyu kiya: Jahan Delhi-specific data nahi mila, wahan honestly ek real (lekin foreign) benchmark use kiya, demo fleet ko synthetic hi rehne diya.

Kaise kiya: Logistic regression → LightGBM (Optuna-tuned) SGCC par; K-means segmentation; Data & Model Card docs/MODEL_RESULTS.md aur API responses se compiled.

Formula / number ka matlab

Theft: $\text{anomaly}(i) = -\text{IF.score}(x_i)$; $\text{cluster}(i) = \text{argmin}_k \|x_i - \mu_k\|^2$

Model Card: $\text{coverage} = (1/N)\ \Sigma\ 1[P10,i \le y_i \le P90,i]$

Viva mein pooche ja sakte hain

Q: Theft ke liye SGCC (China) par kyun train kiya, Delhi data kyun nahi mila?

A: Humne explicitly search kiya ki koi public Delhi ya Indian DISCOM consumer-level smart-meter dataset with theft/anomaly labels maujood hai ya nahi — nahi mila. BRPL/BYPL/TPDDL consumer-level meter data publish nahi karte. SGCC ek widely-used public benchmark hai isi tarah ke problem ke liye, isliye usi par train kiya aur transparently FOREIGN tag diya.

Q: Demo fleet real hai?

A: Nahi, synthetic hai, sirf illustration ke liye jab tak real Delhi meter data available na ho.

Agli slide par: *Ab dekhte hain har page ka benchmark ek table mein.*

Slide 13 — Benchmarks

Slide par kya hai

Table: har page ka benchmark, hamara result, aur source. Niche ek note jo is round ke Solar/Losses updates summarize karta hai.

Bolne wali script (~50 sec)

Yeh slide har page ke result ko uske benchmark ke saamne rakhti hai, ek jagah par. Forecast: persistence 1.44/3.18% ke against humara test MAPE 3.93%, 84.4% coverage. IEX: EAC-PM ka range ₹1.11–9.71 ke against humara real history ₹3.05–6.13/unit. ToD: DERC clause 3(d) se seedha 20%/20%. Solar: do real plants se validate, r^2 0.99 aur 0.21 — Plant 2 ka root cause ab investigate ho chuka hai. Losses: ab BRPL, BYPL aur TPDDL teeno PFC ke primary document se confirm hain. Theft: SGCC paper ke ~0.78 ke against humara 0.7585, no leakage.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Saare benchmarks ek table mein.

Kyu kiya: Examiner ko ek hi jagah dikhe ki kaunsa number measured hai, kaunsa external benchmark se compare hua.

Kaise kiya: docs/BENCHMARKS.md se seedha liya, har row ka apna source.

Viva mein pooche ja sakte hain

Q: Sabse weak result kaunsa hai, aur kyun?

A: Solar Plant 2 (r^2 = 0.21) aur Theft (AUC 0.7585, paper se thoda kam). Dono ko humne khul kar report kiya, chhupaya nahi — Plant 2 ka root cause investigate kiya, aur Theft ka number paper se neeche hai jo leakage ka ulta signal hai.

Agli slide par: *Ab references aur phir closing.*

Slide 14 — References

Slide par kya hai

20 sources: PIB, Delhi SLDC, CSE, EAC-PM, CERC, DERC, IEC 60076-7, textbooks (V.K. Mehta, Kothari & Nagrath, Gönen), Hyndman & Athanasopoulos (FPP3), conformal prediction, Chronos, IEX.

Bolne wali script

Slide par har external number aur equation ka source hai. Data ke liye Delhi SLDC aur Open-Meteo; regulation ke liye CERC DSM 2024 aur DERC; transformer ke liye IEC 60076-7; formulas ke liye V.K. Mehta aur Kothari & Nagrath; aur forecasting ke liye Hyndman ki FPP3 aur conformal prediction ka paper.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Saare sources ek jagah.

Kyu kiya: Har number verify kiya ja sake.

Kaise kiya: Link aur textbook chapter ke saath.

Agli slide par: *Aur ab, closing.*

Slide 15 — Thank You

Slide par kya hai

Thank you, team aur guides ke naam.

Bolne wali script

Toh sir/ma'am, humne ek real Delhi forecast ko uske nateejon tak le jaane ki koshish ki: peak alert, demand ki shape, garmi ka asar, DSM charge, transformer ageing, aur ab IEX cost, ToD billing, solar output, losses aur theft detection bhi usi forecast ke saath, har number par saaf label. Aapke time ke liye dhanyavaad. Ab hum aapke sawaalon ka swagat karte hain.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Closing.

Kyu kiya: Ek line mein project yaad dilana.

Kaise kiya: Sawaalon ke liye invite.

Viva ke tough sawal

Har jawab ka number deck, report ya repo (BENCHMARKS.md, DATA_SOURCES_LOG.md, MODEL_RESULTS.md, DATA_QUALITY.md, engine code) se match karta hai.

1. Naive baseline kya hai aur kyun zaroori hai?

Seasonal-naive ka matlab: kal isi 15-minute block ka load hi aaj ka forecast. Isme koi learning nahi hai. Agar hamara model isse behtar nahi, to model ka koi fayda nahi. Test par naive 5.16% aur LightGBM 3.93% hai, aur model har horizon par naive se behtar hai.

2. LightGBM hi kyun? SARIMAX kyun drop hua?

LightGBM lags, weather aur calendar jaise bahut saare features ek saath le sakta hai, fast hai, aur SHAP se explain hota hai. SARIMAX (hourly) code mein bana hai, lekin validation par order search mein sabse achha result bhi lagbhag 25.3% MAPE tha, aur uska test par score kabhi nahi hua. Isliye woh reported results mein nahi hai.

3. Leakage kya hota hai? Hampel wala bug kaise pakda aur fix kiya?

Leakage matlab model ko training mein aisi jaankari mil jaana jo forecast ke time par hoti hi nahi, jaise future ka data. Pehle Hampel outlier filter ek centred window use karta tha: kisi reading ko outlier kehne ke liye 2 ghante aage tak ki readings dekhta tha, aur sigma par koi floor nahi tha, isliye 11.5% real readings flag ho rahi thi. Ek independent code review mein yeh pakda gaya. Fix: trailing (sirf peeche dekhne wali) 35-minute window, aur sigma par median ke 2% ka floor. Ab sirf 0.1% readings flag hoti hain.

4. Naive ke 5.22% aur 5.16% mein farak kya hai?

Data same hai, rows alag hain. 5.16% poore test set par hai: woh 1,656,240 rows jinke liye kal ka value maujood hai. 5.22% model ladder table mein hai, jo sirf 1,267,749 'common real' rows par hai, yaani woh rows jo har data version mein real reading thi.

5. LF 0.646 aur 0.503 mein farak?

0.646 ek illustrative din hai: 530/820, sirf formula samjhane ke liye. 0.503 poore period (April 2023 – January 2026, 96,398 blocks) ka measured value hai: 4,316/8,582. Dashboard 0.503 dikhata hai.

6. P10–P90 ka target 80% hai aur coverage 84.4% hai. Iska matlab?

P10–P90 ek 80% range hai: theory mein 80% actual values iske andar aani chahiye. Test par 84.4% aayi, yaani band thoda conservative (chauda) hai. Yeh average hai: August mein 73% aur November mein 98%.

7. Test set fully untouched kyun nahi raha?

Pehle LightGBM ki test-set errors dekh kar (woh lambe horizons par naive se haar raha tha) humne target-aligned lag features design kiye. Hyperparameters aur early stopping ne test kabhi nahi dekha, lekin feature design par test ka asar pada. Isliye 3.93% thoda optimistic ho sakta hai; hum yeh khud batate hain.

8. h = 1 par persistence model se behtar kyun hai?

15 minute mein load bahut kam badalta hai (lag-1 block autocorrelation 0.998). Isliye 'agla block = abhi wala block' h = 1 par 1.44% MAPE deta hai, jabki model 3.18%. Hamara ek hi model saare 96 horizons ke liye optimise hua hai.

9. Diwali par error zyada kyun hai?

Diwali (20 October 2025) par day MAPE 8.65% tha aur sirf 49% blocks band mein aaye. Actual load 4-week average se lagbhag 2,000 MW kam tha. Model holiday ko pehchanta hai, lekin training mein holidays bahut kam the aur sirf ek indicator feature hai.

10. Synthetic aur illustrative data kahan use hua aur kyun?

Synthetic: paanch consumer category profiles (residential, commercial, industrial, agricultural, EV), parametric model se bane, ab sirf Theft & Segments ke demo fleet (page 11a) ke liye. Illustrative: DSM ₹ Calculator, Transformer Health aur LF ka 530/820 example. Baaki sab pages real ya validated/foreign data par chalte hain.

11. DSM ₹ figure illustrative kyun hai?

CERC DSM Regulations 2024 mein normal rate grid frequency par depend karta hai. Woh tables encode nahi kiye; humne flat ₹5,000/MWh aur k = 1.5 maana. Isliye ₹3.475 lakh/hr sirf charge ki shape dikhata hai, real bill nahi.

12. 8,748 MW wala peak dataset mein kyun nahi?

Dataset 12 January 2026 par khatam hota hai, aur yeh record 29 June 2026 ka hai. Data ka maximum 8,581.6 MW (19 June 2024) hai. Isliye test window mein March–June ki garmi bhi nahi hai.

13. Overfitting hai ya nahi?

Train, validation aur test MAPE 2.29%, 4.29% aur 3.93% hain (accuracy 97.71%, 95.71%, 96.07%). Gap chhota hai: early stopping ne 3,000 mein se 436 rounds rakhe, aur validation aur test error flat rahi, sirf train error giri. Underfitting bhi nahi: model test par naive se 24% behtar hai, har horizon par.

14. Data clean karne mein kitna badla gaya?

15-minute ke 97,728 blocks mein se 2,047 fill kiye gaye (894 forward-fill, 1,153 profile-fill), 249 outlier flag hue, aur 1,330 blocks khaali chhode. Har filled block par flag hai, kuch bhi chhupa nahi.

15. 96% accuracy kaise nikali?

Accuracy = 100 − MAPE. Test MAPE 3.93% hai, isliye accuracy 96.07% (≈ 96%); train par 97.71%, validation par 95.71%. Yeh classification accuracy nahi hai: har 15-minute block ki galti ko actual load se divide karke average li, phir 100 se ghataya.

16. Solar Plant 2 ka r^2 itna kam (0.21) kyun hai?

Humne chaar hypotheses test ki: timestamp misalignment (shift karke dekha, par shift=0 hi sabse achha fit deta hai), inverter clipping (AC/DC ratio sabhi 22 inverters mein flat ~0.975 hai, koi clipping signature nahi), sensor placement (dataset mein ek hi weather sensor hai jo 22 physically-spread inverters ko cover karta hai — yehi sabse badi wajah lagti hai), aur din-wise pattern (34 dinon ka WAPE continuous spread hai, koi ek discrete faulty din nahi, lekin sabse weak din June ke monsoon-onset ke aas-paas cluster karte hain, jo patchy cloud variability ka sign hai). Koi fixable bug nahi mila, isliye r^2 = 0.208 hi final rakha — yeh real-world sensor-mapping ki seema hai, formula ki galti nahi.

17. BYPL ka AT&C loss kitna hai, aur kaise pata chala?

BYPL ka AT&C loss 7.84% hai, FY2023-24. Pehle yeh secondary source (Power Line) mein nahi mila tha, isliye unconfirmed rakha tha. Is round mein humne PFC ka primary PDF — "Report on Performance of Power Utilities 2023-24" — seedha dhoondh kar padha, Annexure 1.8, page 76. Usi table se BRPL (6.58%) aur TPDDL (5.91%) bhi re-confirm hue, exact match secondary source ke saath.

18. Theft & Segments SGCC (China) par kyun train kiya, Delhi data kyun nahi mila?

Humne explicitly search kiya ki koi public Delhi ya Indian DISCOM consumer-level smart-meter dataset with theft/anomaly labels maujood hai ya nahi — nahi mila. BRPL/BYPL/TPDDL consumer-level meter data publish nahi karte. SGCC (China State Grid) ek widely-used public benchmark hai isi tarah ke problem ke liye, Zheng et al. 2018 paper ke saath, isliye humne usi par train kiya aur transparently FOREIGN tag diya. Demo fleet (page 11a) synthetic hai, jab tak real Delhi meter data available na ho.

19. IEX price ka number kaise verify kiya?

Humara number Kaggle ke real IEX day-ahead market history (2018-2024, 233,760 rows) ka block-of-day average hai. Isko IEX ki apni published price range ke saath compare kiya — humara range (₹3.05–6.13/unit) us range ke andar aata hai. Dhyaan rahe, yeh ek 6.7-saal ka typical-day average hai, kisi specific din ka live price nahi, isliye isko EAC-PM ke one-month snapshot (₹1.11–9.71) se directly compare nahi karte.